

CART-498

Assignment 2: Self-Reflection

This assignment consisted of modifying the poem *The Snow Man* by Wallace Stevens utilizing GPT-2. The first part of the assignment asked to change the last word of each line to the 7th most probable outcome. The second part was to alter the number of probabilities to find the most funny or absurd outcome. I tried many different probabilities before finally settling on the 4800th probable outcome. While altering the probabilities of the GPT-2, I noticed that the model tends to propose words that fit the context of the line with lower probabilities (e.g. 7th most probable outcome). However, this is not always consistent. When I tested a bunch of higher probabilities it gave me outcomes that almost fit better than previous lower probability outcomes. For example, in the P+7 outcome, there is a line that says: “That is blowing in the same bare head”. However, in the P+48 outcome, the same line now says: “That is blowing in the same bare earth”. So, it seems rather inconsistent how the model decides what word would fit best. Yet the further out you get the more absurd the responses become. With P+4800, the model started even using slightly vulgar language which was interesting to see.

The way I would implement a p+7 technique where all nouns are replaced with the 7th highest probable outcome is to first perform a part-of-speech tagging for each line which would identify all the nouns. Then, I would iterate to get each noun token for the respective line. After I have all the tokens, I then feed the prefix up to that noun’s position into GPT-2 and ask for ranked next-word candidates. After this, I must make it select the 7th valid replacement noun. Yet, while I say this, I would employ an LLM to help me write the code and expedite this problem.